NOT MEASUREMENT SENSITIVE

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8 December 1989
SUPERSEDING
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14 July 1983

MILITARY SPECIFICATION

TRUCKS, REFUSE COLLECTION: COMPACTION TYPE, WITH FRONT CONTAINER HOISTING DEVICE; DIESEL ENGINE DRIVEN; 44,500 TO 62,000 POUNDS GVW, 6X4, WITH TILT CAB, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel engine driven, 6-wheel, 4-wheel drive, tilt cab, commercial, compaction type, refuse collection trucks, equipped with a front loading container hoist. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Vehicle classification. The truck shall be one of the following types and classes, as specified (see 6.2 and 6.4):

Type II - 44,500 GVW - 24 cubic yards
Type III - 51,000 GVW - 30 cubic yards
Type IV - 56,000 GVW - 34 cubic yards
Type V - 62,000 GVW - 41 cubic yards

Class B - 4,500 pound hoist
Class C - 6,000 pound hoist
Class D - 8,000 pound hoist
* Class E - 10,000 pound hoist

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	A-A-393	- Extinguisher, Fire, Dry-Chemical (Hand-Portable).
	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.

MILITARY

	MIL-T-704	- Treatment and Painting of Materiel.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
	MIL-R-22827	- Refuse Containers and Modification Kits, Hoisting Units.
	MIL-R-23954	- Refuse Containers, Hoisting Unit; Tilt-over, for Front Loading Compaction Type Trucks.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
*	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
	FED-STD-595	- Colors.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.

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* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards.

(Application for copies of OSHA publications should reference the Code of Federal Regulations 29 CFR Part 1910 and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
ANSI Standards.
Z245.1-84 - Refuse Collecting and Compacting Equipment - Safety Requirements.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code, Section VIII.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM STANDARDS
B117 - Method of Salt Spray (Fog) Testing.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the European Tyre and Rim Technical Organisation publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices
J537 - Storage Batteries.
J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J588 - Turn Signal Lamps.
J589 - Turn Signal Switch.
J682 - Rear Wheel Splash and Stone Throw Protection (DoD adopted).

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J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings for Six- and Eight-bolt Truck Transmission Mounted Power Take-Offs.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.6, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.6, no removal, substitution or alternation of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

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* 3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with MIL-STD-1223 for the military service identified by the procuring activity (see 6.2), except for the following:

(a) The exterior of the complete body unit (see 3.5 through 3.5.7) shall be cleaned in accordance with the manufacturer's standard commercial practice and treated and painted in accordance with type A of MIL-T-704;
(b) The interior of the body shall be painted with rust inhibiting primer, grey in color;
(c) The color of the exterior of the vehicle shall match color chip No. 13538, gloss yellow, of FED-STD-595 for the Navy; color chip No. 16081, gloss grey, for the Air Force; and color chip No. 17875, gloss white, for all other services;
(d) The color of vehicle identification markings shall be gloss black matching color chip No. 17038 for the Navy; reflective white numerals and letters on a red background for the Air Force; and gloss black matching color chip No. 17038 numerals and letters on a reflective white background on the vehicle and front and rear marker plates for all other services;
(e) Rubber seals and hydraulic hoses shall not be painted.

* 3.1.1.2 Rustproofing. When specified (see 6.2), the cab, chassis and complete underside of the refuse collection body including fenders, skirts and wheelwells shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing of these areas in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in the rear axle shall be of the permanent magnet type.

3.1.1.4 Towing devices. Towing devices consisting of two hooks, loops, eyes, or pins or the chassis manufacturer's standard single center mounted eye or pin shall be mounted on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.5 Splash and stone throw protection. Rear wheel splash and stone throw protection in accordance with SAE J682 shall be furnished. Rubber mud flaps or fenders shall be provided to the rear of the front wheels and shall extend down to at least the centerline of the front axle. A metal strip, not less than 1/8 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1-1/4 inch wide by 3/32 inch thick shall be furnished at each bolt. All tilt cabs shall have rubber mud flaps to the rear of the front wheels.

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* 3.1.1.6 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The interior sound level shall not exceed 84 db(A). The interior noise level during the packing and container hoisting cycles shall be within the permissible noise exposure levels of Occupational Safety and Health Standard 1910.95, based on an exposure duration of 8 hours per day. The vehicle exterior sound level shall conform to the Environmental Protection Agency Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; the body and hoist; and a full complement of fuel, lubricants, coolant, hydraulic fluid, and washing solution.

* 3.2.5 Gross vehicle weight. The minimum GVW shall consist of the curb weight; the operator weight (computed at 175 pounds); and a payload distributed as specified in 3.2.6 to provide not less than the minimum GVW specified in table I. In addition, the actual GVW of the vehicle furnished shall exceed the minimum GVW specified in table I when required to provide a payload not less than that specified in table I. Further, the front axle gross axle weight rating (GAWR) shall be not less than 18,000 pounds.

TABLE I. Minimum GVW and payload (pounds).

Type	Minimum GVW	Minimum payload
II	44,500	14,000
III	51,000	18,000
IV	56,000	20,000
V	62,000	24,000

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3.2.6 Weight distribution. The distribution of gross vehicle weight for the purpose of establishing suspension, axle, and tire capacities shall be determined with the appropriate payload (see 3.2.5) distributed throughout the load area. The load area shall be defined as that area rearward of the compaction plate when the compaction plate is in the fully retracted position (see 3.5.2). The compaction area is defined as the area to the rear of the hopper opening. For the purpose of establishing payload weight distribution, the product density of the payload in pounds per cubic foot in the compaction area shall be considered to be twice the product density of the payload in the hopper area. Loads imposed on vehicular components for loaded over-the-road conditions shall be determined with hoist arms in the travel position and with no container load engaged.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity.

3.2.8 Cab-to-trunnion. The cab-to-trunnion dimension shall conform to that recommended by the body and hoist manufacturer for conformance to all the requirements of this specification.

* 3.2.9 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors, air horns and tires shall be not more than 96 inches. The width over the tires shall be:

(a) not more than 100 inches for axles rated up to and including 46,000 pounds
(b) not more than 102 inches for axles rated at over 46,000 pounds and up to and including 58,000 pounds.
(c) not more than 104 inches for axles rated at over 58,000 pounds.

* 3.2.10 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.11 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the truck loaded to specified GVW.

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3.3.1.1 High speed gradeability. The vehicle shall ascend a 0.50 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed not greater than 2-1/2 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine with not less than six cylinders. The engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristic specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

* 3.4.1.2 Oil filter(s). Full flow type oil filter(s) shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

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* 3.4.1.4 Cooling system. The cooling system shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

* 3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. Control shall include complete thermostatic control of all coolant flow through the radiator.

3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

* 3.4.1.7 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. The heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. A carrier for the connecting cable shall be mounted within the cab and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(a) A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed.
(b) An immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening.
(c) The battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than plus 80°F.

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(d) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

3.4.2 Electrical systems. Electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. The starting system shall be 12- or 24-volt with 12-volt direct current (dc) lighting system, and with not less than a 60-ampere alternator. Engine starting equipment shall include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether systems shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ampere ratings at 0°F shall be not less than specified in table II. Reserve capacity and cold cranking ratings shall be measured in accordance with SAE J537. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE II. Batteries.

	Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
	Diesel engine (less than 220 gross horsepower)	320	1,200
*	Diesel engine (220 gross horsepower and over)	480	1,740

* 3.4.2.3 Lighting. The electrical system shall be 12 volts. All vehicle lights and reflectors shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on rub rails or vehicle bumpers (unless recessed and fully protected).

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* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible and audible type flash indicator.

* 3.4.2.5 Back-up lights, warning signal and video camera system. The vehicle shall be provided with back-up lights and with an audible, pulsating, signaling device (electric or mechanical) to caution personnel when the vehicle is in reverse gear operation. When specified (see 6.2), the vehicle shall be furnished with a video camera system. The camera shall be mounted on the rear of the vehicle, and a monitor shall be mounted on the dash. The system shall clearly show objects to the rear of the vehicle. The monitor shall have a grid screen indicating distance of objects and a day or night switch to adjust brightness for changing conditions for night driving. The system shall be activated whenever the transmission is in reverse gear.

3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. A dry type air cleaner with dash mounted restriction gage shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall have not less than 50 gallons (43 for vehicles destined for California) total capacity. When more than one tank is furnished, means shall be provided to assure an equalized fuel level in both tanks.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Vertical exhaust mufflers capable of being reached easily by personnel entering or leaving either side of the cab shall be provided with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap.

3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

* 3.4.5 Automatic or semiautomatic transmission. The vehicle shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque

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convertor and not less than five forward gear ratios. The normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with a power takeoff opening conforming to SAE J704. A neutral-to-drive shift inhibitor shall be provided, preventing shifts from neutral to drive while the engine is operating at speeds greater than 900 revolutions per minute.

3.4.5.1 Power takeoff. A power takeoff shall be provided. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the cab accessible to the seated driver. The power takeoff shall be air controlled. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard, reinforced frame, or a heavy duty frame, for the type and class vehicle furnished. The reinforced frame shall have frame reinforcement extending at least from the rear of the front suspension, rear hanger bracket to the bogie trunnion mounting bracket. Reinforcement in the form of fishplating is not acceptable. Both the reinforced frame and the heavy duty frame shall have sufficient resisting bending moment to provide the structural strength required to at least equal the loads imposed with the vehicle loaded to provide specified GVW.

* 3.4.8 Suspension. The vehicle shall be equipped with a suspension system having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle both empty and loaded with the appropriate payload as specified in 3.2.5 and 3.2.6. In addition, the front suspension system rating shall be not less than 18,000 pounds. When the spring capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers on the front axle if offered as standard or optional equipment on the chassis model furnished. With the fully loaded truck standing still, the front suspension system shall be capable of withstanding, without permanent distortion, the loads imposed during hoisting operations by a container loaded to hoist capacity.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground when the vehicle is both empty and loaded with the appropriate payload as specified in 3.2.5 and 3.2.6. In addition, the front axle rating shall be not less than 18,000 pounds. With the fully loaded truck standing still, the front axle shall be capable of withstanding, without failure, the loads imposed during hoisting operations by a container loaded to hoist capacity.

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* 3.4.9.1 Rear bogie. The rear bogie shall be of the four-wheel type, complete with axles, springs, torque rods, and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. Lockout control shall be located in the truck cab and shall be equipped with a warning light to indicate when the lockout is engaged.

* 3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished on front axles and on rear tandem axles rated at 52,000 pounds or less. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of the steel belted radial ply type. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with vehicle both empty and loaded with the appropriate payload as specified in 3.2.5 and 3.2.6. In addition, front tire total capacity shall be not less than 18,000 pounds. With the truck both fully loaded and empty and standing still, the front tires shall be capable of withstanding, without failure, the loads imposed during hoisting operations with a container loaded to hoist capacity. Tires and tire size designation systems shall conform to Tire and Rim Association or to The European Tyre and Rim Technical Organisation recommendations.

3.4.10.2 Inner tubes. For tube type tires, inner tubes shall be of the heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), and 393.45 through 393.52. Brake linings shall be of nonasbestos material on front axles and on tandem rear axles rated at 46,000 pounds or less.

3.4.11.1 Air brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute
(b) Air storage reservoir(s) with not less than a 3,360-cubic inch total capacity; each tank equipped with a drain, and with safety and check valves between the compressor and the last reservoir tank

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(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air-pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) For vehicles with rear axles rated at 46,000 pounds or less automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes on all axles. Vehicles without this feature on one axle (rear axles rated at more than 46,000 pounds) shall not have it on any axle.

* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when a spring applied emergency brake is furnished), and additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. A full width tilt type cab accommodating the driver and at least one passenger, with a tilting and locking mechanism and a bumper-to-back-of-cab dimension of not more than 87 inches, shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least one door equipped with an external key-operated lock. Drip rails shall be provided above the cab doors. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab. A steel bar to act as a guard shall be attached to the front of the cab. The guard shall extend from the top of the cab to below the windshield vertically and shall be located as near to the center of the windshield as practicable. Manufacturer's standard seat belts for each seating space shall be installed. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be

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furnished. Provisions for cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and individual passenger seat shall be provided. When specified (see 6.2), the driver's seat shall be the air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. A periscope type mirror screen shall be installed in the ceiling of the cab to enable the operator to view the container during all stages of the container dumping operations.

3.4.12.2 Vision. The cab shall provide the operator with an unobstructed view of the left side of the two-point hookup container or in the case of a single-point hookup, at least the upper 10 inches of the hoist hookup, when the truck is at the point of contact with the container. These conditions shall be met with the operator seated in the cab in the normal operating position and with the container resting on the ground at the truck wheel level. This requirement shall be met with containers which are not provided with casters.

3.4.13 Steering. Power steering shall be furnished, and shall enable the operator to turn the front wheels to the extreme right or left while the vehicle, both empty and loaded to rated GVW, is parked on a dry, paved surface.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by air or electric motor(s).

3.4.15 Bumper. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided.

3.4.16 Tool compartment. A covered tool compartment with provision for locking with a padlock shall be installed. The compartment shall be of adequate capacity to contain hand tools, antiskid chains, and emergency reflective triangles. If the compartment is installed outside the cab, it shall be weatherproof.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to windshield defroster louvers. The heater shall be complete with a blower and mounted controls convenient to the driver.

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3.4.18 Controls and operating mechanisms. All controls and operating mechanisms inside the cab shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Controls requiring disconnection in order to tilt the cab shall be provided with automatic disconnect/connect as the cab is tilted. Controls requiring disconnection in order to tilt the cab shall be provided with automatic disconnect/connect as the cab is tilted. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Openings in the cab, for the purpose of installing control levers, shall be sealed with a boot or other flexible type cover to prevent roadwash and dirt from entering the cab. All instruments inside the cab shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. The chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, voltmeter, or charging indicator
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light
* (e) Engine temperature gage or high coolant temperature or low coolant level red indicator light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Tachometer
(i) Ash receptacle
(j) Dual sunvisors
(k) Driver's compartment ventilator other than window.

3.4.20 Horn. An electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

3.4.21 Engine hour meter. An hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to assure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

* 3.4.22 Mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall have not less than two supports. The mirrors shall be of the combination type having flat and convex areas. Each combination mirror assembly shall be adjustable and shall have at least 50 square inches of flat reflective area and a convex surface having at least 20 square inches of reflective surface and a uniform radius of curvature of not less than 35 inches and not more than 65 inches.

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3.4.22.1 Mirror mounting. Outside rearview mirrors shall be mounted in a manner which provides the driver an unobstructed view along the entire length of each side of the body. When the loading arms of the hoist do not clear the mirrors during operation, the mirrors shall be provided with an automatic retracting device. The mounting shall provide sufficient adjustment to accommodate all drivers.

* 3.4.23 Fire extinguisher. A 5-pound dry chemical type fire extinguisher, conforming to Federal Motor Carrier Safety Regulation 393.95(a), shall be mounted convenient to the driver. The fire extinguisher shall have an Underwriter's Laboratory rating of 2A:10B:C or more, shall include a metal safety pin and wire, and shall conform to type I, class 1, size 5 of A-A-393. The fire extinguisher shall permit visual determination as to whether it is fully charged.

* 3.4.24 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard air conditioning system. The air conditioning system shall include tinted windshield and tinted glass.

3.5 Body unit. The body unit shall be of the compaction type packer body and front loading type hoist. The unit shall be designed so that when installed on the truck chassis it can be operated satisfactorily without the use of truck counterweights, jacks, or outriggers. The hoist shall be capable of engaging refuse containers at the front of the truck, lifting the loaded containers, and dumping the container refuse through the hopper opening in the top of the packer body. The packer body shall be capable of compacting the refuse and ejecting it through the rear of the body. The complete operation of hoisting, compacting, and ejecting shall be capable of being performed by a single operator. The operator shall not require any outside assistance or be required to leave the truck cab except to release or fasten the body rear door lock.

3.5.1 Packer body. The packer body shall be of general welded construction. The sides and top shall be fabricated of not less than 12 gage sheet steel. The floor shall be fabricated of not less than 10 gage high tensile steel or 7 gage carbon steel. The sides, top and floor shall be reinforced to withstand continuous operation at maximum imposed loads without evidence of deformation or excessive wear. The interior of the body shall have a smooth, easily cleanable surface and shall be free of pockets and restrictions that might retard either the loading or unloading operations. Two rows of retainer teeth shall be furnished inside the roof of the body. The body shall retain all refuse material, including ashes and liquid-saturated garbage, without spillage or liquid leakage during refuse collection and transport operations. The lower portion of the entire body shall be liquid-tight to a height of not less than 5 inches. All exterior box bracing shall be continuously welded to prevent bleeding of rust from inside the members onto the exterior of the vehicle. The span of unsupported floor between the underfloor crossmembers shall be not more than 6 inches

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when 10 gage high tensile steel or 7 gage carbon steel is used for the entire floor. The span of unsupported floor between the underfloor crossmembers shall be the manufacturer's standard when 3/16-inch or thicker high tensile steel is used for the floor. The body shall be mounted to the truck chassis using the manufacturer's standard direct mount or wood stringer sills and tiedown angles between the body and the chassis frame.

* 3.5.1.1 Capacity. The refuse body shall have a compaction volume of not less than 24 cubic yards for type II, 30 cubic yards for type III, 34 cubic yards for type IV and 41 cubic yards for type V.

3.5.1.2 Hopper. The loading hopper shall have an opening at the top of the body. The hopper capacity, opening size, and general configuration shall assure complete discharge of refuse from each container specified in 3.5.3.1. The opening shall be provided with a hydraulically operated cover and with wind guards. A warning device shall be installed in the cab to alert the operator when the hopper cover starts to rise. The cover shall open and close automatically with the movement of the container hoist or packer plate to assure that:

(a) Containers can be placed in a dump position only when the cover is fully open;
(b) Cover is completely closed and secured when the hoist is in the vehicle travel position;
(c) Cover operation will not interfere with the hoist and containers.

3.5.1.2.1 Wind guard. Metal hopper extensions to act as wind guards shall be provided at the hopper sides to minimize the possibility of spillage that may be encountered on windy days when containers are being dumped. Wind guards shall be at least the full length of the hopper opening and a minimum of 24 inches above the top of the body. The wind guards shall provide not less than 50 square feet of protection.

3.5.1.3 Side access door. The forward end of the body shall be equipped on at least one side with a side opening not less than 28 inches wide and with an open area of not less than 900 square inches. Side openings shall be covered by a sliding door held in closed position by a lever type, spring-loaded latch. When the hoist arms are so designed that they do not interfere with the side door or if the hoist arms are inoperative with the side door open, hinged type doors capable of being secured in both open and closed positions may be furnished.

* 3.5.1.4 Rear ejection opening. The rear opening shall be the full body width and height. The opening shall be fitted with a reinforced steel door and synthetic rubber seals at the sides and bottom. The door shall be hinged at the top and provided with a hydraulic cylinder to open the door and keep the door in an approximately horizontal plane while the body is being unloaded. The door shall be held in a closed position by spring loaded or

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adjustable cam-over-center latches operated by a single lever located to the side of the vehicle or operated by power from controls inside the cab. An alarm, in accordance with ANSI Z245.1, shall be furnished to warn the operator that the door is open or is not locked.

3.5.2 Compacting and ejection plate. The compacting plate shall permit smooth operation without bypassing refuse during operation, and shall ensure packing refuse in all body areas to the rear of plate. The packer plate assembly shall be powered for both compaction, unloading and retraction by a hydraulic telescopic action cylinder exerting at the end of its first stage not less than 55,000 pounds of horizontal thrust. The ejection plate shall travel the full length of the body. The system shall permit the hoisting mechanism to dump refuse only when the packer plate is in full forward position. When in full forward position, the upper edge of the packer plate shall be forward of the front edge of the hopper opening. The compaction and ejection plate shall completely unload the body without outside assistance. A means shall be provided to automatically prevent movement of the compaction plate when the side access door(s) are open. A warning light shall be furnished in the cab to indicate that the packer is in its compaction cycle.

3.5.2.1 Limit and overload devices. Limit devices shall be incorporated to prevent overtravel of the packing device at both ends of its travel. Overload devices shall be incorporated in all hydraulic circuits to prevent damage to any operating mechanism. Adjustable overload devices shall be set and sealed.

* 3.5.3 Container hoisting mechanism. The hoisting mechanism shall be hydraulically operated and provided with check valves to prevent uncontrolled descent due to hydraulic failure. The hoisting mechanism shall be capable of lifting and dumping a gross load (container and refuse) of not less than 4,500 pounds for class B, 6,000 pounds for class C and 8,000 pounds for class D. The hoist capacity rating shall be based on the gross load located 30 inches forward of the position of the rear container wall with the container in pickup position. The unit shall be capable of engaging, lifting, dumping, returning and disengaging the containers from the ground and from loading docks not less than 48 inches above ground level. There shall be no possibility of the containers striking the cab at any time during normal hoisting and lowering operations. Stationary mechanical cushioned stops to form a rest for the hoist arms when the arms are in the maximum dumping position shall be provided. Hoist arm forks/tines shall have the capability to retract or fold vertically (or near vertical) for vehicle travel. When specified (see 6.2), rests shall be provided to hold the hoist arms in a position suitable for carrying a container being hand-loaded.

3.5.3.1 Containers utilized. The hoisting unit shall be capable of handling containers of the applicable class in any size and type classified in MIL-R-23954 without modification of the containers. When specified (see 6.2), to supplement an existing refuse collection system, the hoisting unit

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shall be compatible with the existing system and capable of handling a specific class of container (as classified in MIL-R-23954) without modification of the container. When specified (see 6.2), the hoisting unit shall be capable of handling the specified type and size of drop bottom containers covered by MIL-R-22827 modified to be used with the front hoist system.

3.5.3.2 Controls and safety. The hoisting unit controls shall be located in the cab. All controls for hoisting shall be of the deadman type. The position of the container and hoist arms shall be under the control of the operator at all times, so that the operator may reverse, stop, or start the movement at any point in the cycle. A safety feature shall be designed into the hoisting system to minimize possibility of injury to operating personnel from the hoist arms. The safety feature shall be such that the hoist arms, in all positions, shall be clear of the cab doors to the extent that there shall be no interference with the entrance or exit of the driver or passengers or interference with the cab doors opening to any position.

3.5.4 Sump. The body shall be provided with a sump tank of not less than 16-gallon capacity. The bottom of the body shall be designed to drain liquids from refuse within the body, into the sump tank. The tank shall be provided with a drain valve and a gasketed hinge cap. The design shall permit thorough flushing of the tank, and liquid lead-ins with a water hose stream.

3.5.5 Wash-out system. A reservoir of not less than 80-gallon capacity shall be installed for washing and disinfecting the interior of the body and refuse containers. The reservoir shall include a fill valve or opening with filler neck and a discharge outlet. The fill opening shall be covered with a cap that has a handle or T-bar to enable closing and opening by hand. When the reservoir furnished is the pressurized type, a caution plate shall be provided to warn that pressure must be relieved before opening cap and the cap shall be of the pressure type. The washing solution shall be discharged under pressure from either compressed air or from an equivalent mechanical pump. When compressed air is utilized to provide the discharge pressure, the reservoir shall be a pressure vessel designed for an operating pressure of not less than 100 pounds per square inch gage. The air system shall be furnished with a safety relief valve, a dial pressure gage, an air charging valve, and a bleed valve. The bleed valve shall be capable of bleeding off the system pressure in not more than five minutes. The air operated reservoir shall be connected to the truck chassis compressed air system, and shall incorporate an air shutoff valve between the compressed air source and the liquid reservoir. The mechanical system shall include a reservoir vent to assure unrestricted discharge of the washing solution. Pressure vessels shall be ASME inspected, stamped and registered with the board of Pressure Vessel Inspection. The pressure vessel shall be in accordance with ASME Boiler and Pressure Vessel Code, Section VIII.

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3.5.5.1 Hose. The tank discharge outlet shall be provided with 50 feet of rubber hose fitted with a spring-actuated shutoff spray nozzle. The hose shall be capable of withstanding the wash-out system pressure. A hose rack shall be provided on the body for the storage and carrying of the hose.

3.5.6 Operating mechanism. Mechanisms for lifting and dumping refuse from containers and for compacting refuse shall be hydraulically operated. The mechanisms shall be actuated by a heavy duty, antifriction, gear or vane type pump. The pump shall be operated by a power takeoff attached to the transmission. All mechanical structures shall be capable of withstanding a 100 percent overload. The hydraulic system shall be capable of withstanding a 10 percent increase in pressure over the system's maximum operating pressure. The lifting mechanism shall be installed so it can be removed to facilitate repairs or service.

3.5.6.1 Hydraulic system. Relief valves shall be provided in the hydraulic system to prevent excessive hydraulic pressure. A hydraulic T-fitting, or equivalent, for inserting a pressure gage shall be provided in the line to facilitate checking the line pressure. A hydraulic fluid filter and reservoir shall be furnished and provided with a fluid level indicator. Hydraulic lines, hoses, and fittings shall meet or exceed SAE recommended practices and standards. The hydraulic ram piston rods shall be stainless steel or shall be chrome-plated full length with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM Designation B117. The connecting ends shall be treated in the best commercial practice.

* 3.5.6.1.1 Hydraulic fluid. Unless otherwise specified, the manufacturer's standard or recommended hydraulic fluid shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. A permanent decal located near the hydraulic system fill port shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-83282 ONLY."

3.5.7 Lubrication. All moving parts of the equipment normally requiring lubrication shall have means provided for that lubrication. Pressure grease fittings shall be of the hydraulic type with a surface check ball and when applied to sealed bearings shall have a pressure release to protect the seals.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle b the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alinement of lights; adjustment of the engine, electrical and brake systems; filling and charging of the battery; alinement of front wheels; inflation of all tires; complete lubrication of the chassis, engine and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

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3.7 Workmanship. Defective components or parts and assemblies, which have been repaired or modified to overcome deficiencies, shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

3.7.1 Body and hoist welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welding rods shall be compatible to the metal being welded. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. The following defects shall not be acceptable:

(a) Welds which are ground flush with metal
(b) Slug or plug welding, except for floor
(c) Undersize fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments, and vertical down welding
(d) Unremoved slag and spatter
(e) Cracks
(f) Spot, tack, or intermittent welds for strength.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's

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inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

* 4.3 First production vehicle inspection. The first vehicle of each type produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Records. The documentation used for the first production vehicle inspection shall be reflective of that which will be used by the contractor in conjunction with his quality control system in any further production under the contract. Aside from documenting the basic inspection, the contractor shall have any supporting objective evidence readily available for review by the Government. The objective evidence can include, but is not limited to, such items as purchase orders, certifications, test reports, in-process inspection records, specimens and similar data which may aid in verifying compliance.

* 4.3.2 Hoisting unit. The truck hoisting mechanism shall be tested to ensure complete functional operation including the engagement, hoisting and dumping of containers fully loaded with refuse and the lowering and disengagement of the empty containers. Tests shall be made using the applicable class and a type A, size 8 container as specified in MIL-R-23954. The contractor shall provide the container(s) to be used for the test. The testing shall include random reversal of hoisting-lowering operations from a loading dock and from ground level and a complete checkout of the controls and safety systems. The test shall continue until the truck is filled to capacity.

* 4.3.3 Compaction unit. The truck compaction unit shall be tested using refuse to ensure complete functional operation of the compacting ram and to ensure complete ejection of the compacted refuse.

* 4.3.4 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and the distribution of curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds, and the payload and weight distribution required by 3.2.5 and 3.2.6. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

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* 4.3.5 Road test. The vehicle shall be driven for a distance of not less than 200 miles over a variety of road surfaces at allowable traffic speeds up to at least 50 mph. The roads shall include typical curves, bumps, turns and stops as may be encountered in normal driving. At least fifty percent (100 miles) of the road test shall be accomplished with a full load of refuse, shall include five sudden stops and shall include ten miles of operation on gravel or otherwise unimproved road surfaces.

* 4.3.5.1 Examination. When testing is completed, the vehicle shall be cleaned and then examined for evidence of distortion, malfunction, damage and leakage. The contractor shall be responsible for correcting defects discovered during operation and for taking corrective action to prevent the recurrence of such defects in any subsequent vehicles.

* 4.3.5.2 Watertightness. After the visual examination is complete, the lower portion of the body shall be filled with water to a depth of approximately 5 inches and allowed to sit for not less than thirty minutes to validate the liquid-tight requirements of 3.5.1.

* 4.3.6 Production sample. Upon acceptance of the first production vehicle it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped under the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

* 4.3.7 Failure of first production vehicle. The first production vehicle shall successfully complete the entire road test and shall meet all requirements of the contract. Failure to successfully complete the road test or to meet requirements shall be cause for the Government to refuse acceptance of the vehicle pending corrective action and reinspection or retest. The necessity for replacement of any major component during the road test may mandate a complete retest at the discretion of the Government. Rejection of the road test vehicle shall be for damage or deficiencies including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalignment of wheels, drivetrain or frame
(d) Vibration due to type of construction or mounting
(e) Evidence of abnormal tire wear
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicle, components or accessories
(h) Loose mounting of parts or accessories
(i) Fuel or hydraulic leakage.

Any deficiency may, at the Government's direction, be cause for retest of any function which is deemed by the Government to be related to the deficiency.

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* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, ensure that the vehicles conform to the physical dimensional and performance requirements specified herein. Each unit shall receive a complete functional check without refuse and at least a 10 mile road test prior to being presented to the Government. At the point of final acceptance, the contractor shall have available relevant records indicating that all inspections, tests, servicing and adjustments have been made.

* 4.5 Welding. The selection of weld type and design are strictly within the purview of the contractor as long as they meet the performance requirements, unless otherwise restricted by the contract. The Government does reserve the right to review the established procedures and their method of conveyance between the contractor's design element and the production area. Such items as drawings, sketches, procedures, specimens and the like may be requested at the time of inspection to evaluate internal controls and compliance. Conformance to 3.7.1 shall be evaluated.

* 4.6 Workmanship. Workmanship shall be evaluated for compliance to 3.7 and normally anticipated manufacturing and fabrication techniques required to fulfill a contract for this type of product. All workmanship and manufacturing techniques shall meet the highest commercial standards. Fabrication steps shall result in exacting fits free of burrs, chips, kinks and similar defects. Reworked items shall be acceptable only if they result in like-original-condition items and fit. The final finish process shall be inspected by the contractor during sanding, priming, painting and repainting to assure there are no runs, sags, orange peel or other rough surfaces and that the procedure meets or exceeds common industry practice and anticipated coating protection for the expected environment and operating conditions. Proper precleaning, preparation and masking shall be such as to prevent overspray on unpainted surfaces such as moldings, trim and accessories. The entire vehicle shall be critically inspected to determine compliance with this specification, related industry practices, common quality standards and workmanship standards for this type of unit.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The refuse truck is intended for nontactical use in front-loading (from special containers), compacting, transporting, and dumping miscellaneous refuse generated at military installations.

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* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of vehicle required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for painting and marking (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Tropical rustproofing, if required (see 3.1.1.2).
* (g) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(h) Power plant heaters, if required (see 3.4.1.7).
(i) If a battery heater is not required with power plant heaters (see 3.4.1.7).
(j) Silicone rubber radiator and heater hoses, if required (see 3.4.1.8).
* (k) Rear view video camera system, if required (see 3.4.2.5).
(l) Disc type wheels, if required (see 3.4.10).
(m) Brake controls for use from a towing vehicle, if required (see 3.4.11.2).
(n) Individual driver's and passenger seats, if required (see 3.4.12).
* (o) Air ride driver's seat, if required (see 3.4.12).
(p) Air operated horn, if required (see 3.4.20).
* (q) Air conditioning, if required (see 3.4.24).
(r) Hoist rests, if required (see 3.5.3).
(s) Make of existing equipment and class of container, if required to supplement system (see 3.5.3.1).
(t) Type and size of modified drop bottom containers, if required to handle (see 3.5.3.1).
(u) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.5.6.1.1).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computation for low speed and maximum geared speed will be required by the contractor. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type and class. Unless other conditions are cited in the contract, computations shall be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.3.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

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Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613.
Table 4	- Area Factor. 0.240.
Table 5	- Velocity Factor. 250.0.
Table 6	- Altitude Factor. 1.00.
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher 1000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75.
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

6.4 Change in classification. In accordance with interservice agreement during previous coordination of this specification, type I, 39,500 GVW, 24 cubic yards and class A, 3000 pound hoist were deleted.

* 6.5 Subject term (key word) listing.
Commercial vehicle
Compaction unit
Nontactical vehicle
Refuse
Video camera.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

Custodians:
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Navy - YD
Air Force - 99

Review activities:
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Preparing activity:
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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-T-46748G TRUCKS, REFUSE COLLECTION: Compaction Type Body, W/Front Container Hoisting Device; DED; 44,500 To 62,000 Pounds GVW, W/Tilt Cab, Conv.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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